🔗 Topic 1 at a glance: which bones can we trust?

- **Superimposition** lines up two X-rays on structures that do not change; whatever still differs is growth or treatment.
- **Gold standard:** Björk's metal implants in 2D; bone plates and screws in 3D today.
- **Maxilla:** register on the **zygomatic process (key ridge)**. The palate sinks and its plane tilts, so palatal "best fit" at ANS hides **30–50 %** of eruption.
- **Mandible:** register on the **chin and inner symphysis**; the third-molar crypt and the mandibular canal move with growth.

An Investigation of Maxillary Superimposition Techniques Using Metallic Implants

by **Doppel, Damon, Joondeph and Little** · Am J Orthod Dentofacial Orthop 1994;105(2):161–168 · University of Washington

🎯 Study purpose

To find **stable anatomic landmarks in the maxilla** that can be used reliably for cephalometric superimposition, judged against **metallic implants (gold standard)**.

🔬 Study design

- **50 patients** (23 M, 27 F), chosen from about **700** implant records (Univ. of Washington, 1967–1975); most had lost or misaligned implants.
- Most were treated (edgewise, headgear, Class II or III elastics; extraction or not). Retrospective, descriptive: no statistical tests.
- Age at T1 **11.9 ± 1.4 y** (range 8.7–16.0), at T2 **16.0 ± 1.7 y** (11.9–20.3); interval **3.0–6.4 y** (mean **4.2 ± 0.96 y**) (Table I).
- **Implant sites (Björk):** (1) below ANS (anterior), (2) zygomatic process (lateral), (3) palate–alveolar border medial to the first molar.
- **Selection:** all three implants in perfect orientation, films ≥3 years apart, high quality, same cephalostat (facing right), same head tilt and rotation.
- **Why head position matters:** Julius found the zygomatic implant shifted up to **3 mm** with head rotation; a patient fixed in a cephalostat can still rotate the head **3°** and tip it **5°**.
- **Head orientation checked in 4 areas:** inferior and posterior borders of the mandible, orbits (same relative position), infratemporal fossae (same distance apart).
- One operator; 10 films traced twice 2 weeks apart: ±0.25 mm, total tracing error **0.5 mm** (Thurow's limit).
- Films superimposed on the implants; each structure's maximum drift from perfect fit measured with digital calipers (0.01 mm); curved surfaces averaged over several points.

📍 Landmarks evaluated (17, Methods list)

- **Cranial base:** clinoid process (anterior and posterior aspects), anterior cranial base (S-N line), ethmoid triad, nasion.
- **Maxillary skeletal:** zygomatic process ("key ridge", anterior and posterior aspects), lateral border of the orbit, orbital floor, infraorbital foramen, PTM, palate (superior and inferior borders, internal structure), ANS, A point, PNS.
- **Dental:** central incisor, first permanent molar. **Other:** radiopacities that appeared on both films (the nasal floor is not one of the 17).
- Table II also has 17 rows, but not the same 17: the zygomatic process, the palate borders, PTM and ANS each take two rows; the teeth, nasion and radiopacities have none.

📊 Key results: drift from implant superimposition (Table II)

Structure	N	Mean ± SD, mm	Note
Internal palatal structure	50	0.85 ± 0.43	Best number, poor fit by eye: limited value
Anterior zygomatic process	40	**0.95 ± 0.57**	Reliable; traceable in 80 %; direction varies (27.5 % forward, 35 % back, 37.5 % none)
Posterior zygomatic process	50	**0.97 ± 0.43**	Reliable; traceable in 100 %; always moves back (apposition)
Palate, superior / inferior border	50	1.05 ± 0.52 / 1.14 ± 0.60	"Next in reliability" in Results; not reliable in Conclusions
Lateral border of orbit	50	1.27 ± 0.72	Too low for clinical use
ANS, vertical / horizontal	50	1.29 ± 0.86 / 1.43 ± 1.14	Not reliable; palatal rotation and remodelling
Orbital floor	47	1.40 ± 0.98	Rises with growth (apposition)
Infraorbital foramen	34	1.67 ± 1.31	Not reliable; text says traced in 32 %, Table II has 34 of 50
A point	50	2.13 ± 1.63	Not reliable; moves with tooth movement and treatment forces
PNS	27	2.27 ± 1.75	Not reliable; visible in about half (text 52 %); lengthens, moves vertically
PTM, vertical / horizontal	50	2.39 ± 1.69 / 1.52 ± 0.99	Very inaccurate
Clinoid / cranial base / ethmoid	50	2.64 / 2.52 / 2.52	Not maxillary: the maxilla moves against the cranial base (our reading)

Smaller drift = more reliable. The two zygomatic surfaces are **practically equal (0.95 vs 0.97 mm)**: neither is "the most stable" alone; the posterior is traceable in every film with the smaller SD.

🔄 Rotational changes against the implant line (Table III)

Plane	N	Mean	SD	Range
Palatal plane (ANS–PNS)	50	+0.92°	2.10°	−5° to +8°
S-N line (transferred, Björk)	50	−0.56°	2.19°	−6° to +9°

Positive = downward and backward (Results text; Table III prints minimum and maximum the wrong way round). ANS–PNS should not move against the implants, yet spans 13° (derived). The maxilla rotates against the cranial base, and the palatal plane rotates inside the maxilla, differently in each patient.

⚠️ Discussion and limitations

- **Anterior zygomatic surface:** no change in some, resorptive in others (Kurihara, Enlow and Rangel), forward in others (transverse rotation of the maxillae). Björk and Skieller matched it almost exactly in all 9 cases, registering the implant line at the lateral implant.
- **Internal palate:** small numbers, but the contours did not superimpose well by eye.

- **Maxillary rotation:** Björk and Skieller (21 untreated, 6-year intervals) found compensatory alveolar apposition and nasal-floor resorption varying with rotation; treatment adds variability here.
- **Limitations (authors):** head-position change may mimic implant fit (no bilateral implants); with closely spaced implants, drift grows with distance from them; curved-surface fit judged by eye. Our note: the 1.5 : 1 ratio is stated, not tabulated.

🧠 Main conclusions

1. The **anterior and posterior zygomatic contours** matched the implants closely (about 1 mm) and were the best usable structures; the anterior varied (none, slightly forward, slightly back), the posterior always showed apposition.
2. **Vertical remodelling:** the orbital floor rises more than the palatal plane lowers, 1.5 : 1 (in agreement with Björk and Skieller, and with Nielsen).
3. The maxilla rotates variably against the cranial base.
4. The palatal plane rotates variably within the maxilla.
5. Internal palate of limited value.
6. **Infraorbital foramen, PTM, PNS, ANS, A point and the palate borders are not reliable.**
7. No maxillary area is completely reliable; register on both zygomatic contours. (The paper numbers six conclusions; this one comes from the Discussion and the abstract.)

🦷 Clinical implications

- **Best approximation:** register on the **anterior and posterior zygomatic contours**, letting the orbital floor rise 1.5 × the palate's lowering (misprinted "1:5 to 1" on page 165).
- The registration changes the clinical conclusion: with ANS or PTM registration the implants themselves appear to move and the tooth movement is misread (one case, Figs 1–6, against the implants in Fig 7, a case close to the mean). Baumrind found that ANS–PNS superimposition masks downward remodelling.

🔀 Illustrative comparison of methods (Figs 1–7)

Fig	Registration	Implants appear to move	Teeth	Occlusal plane
1	Palatal plane at ANS	Up and back	Incisors look retracted, molars descend	Up and forward
2	Palatal plane at PTM	Up and forward	Labially moved, little descent	Somewhat up, forward
3	Infratemporal fossa, posterior palate	Down and back	Incisors erupt down and lingually; molars vertical	Markedly down, back
4	Internal palate best fit	As Fig 1	Incisors lingual, molars vertical	Slightly up, forward
5	Anterior zygomatic border (inferior point)	Slightly down	Descend; molars slightly mesial	Very slightly up
6	Posterior zygomatic, on internal palate	Slightly up, forward	Molars forward; incisors erupt less	Not described
7	Implants (reference)	None	Case close to the mean findings	Reference

📝 Background the paper builds on (Introduction)

- **Björk and Skieller:** the anterior–lateral implant distance shortens apparently with growth: the two maxillae rotate transversely with midpalatal-suture apposition, **more posteriorly** (in effect, the palate opens like a hinge).
- They found the **anterior zygomatic surface strikingly stable sagittally in all 9 cases**; orbital-floor apposition to nasal-floor resorption **3 : 2**; the **posterior zygomatic surface appositional**, most at its inferior part.
- **Their method:** draw S-N on film 1, transfer it on the anterior cranial fossa and anterior wall of sella, then superimpose the anterior zygomatic contour. Shows rotation well, but **alveolar height growth is hard to evaluate** (no vertical reference).
- **Moss and Greenberg** (47 Eskimo skulls) registered on the infraorbital foramen, judged constant; **Björk and Skieller disputed this**: it follows the orbital floor and can be moved by orthodontic forces.
- **Julius (21 untreated implant subjects):** (1) posterior and key-ridge zygomatic surfaces parallel, about the same place; (2) anterior cranial base and cribriform plate parallel; (3) infraorbital foramen relocates forward and vertically; (4) palatal plane rotates down and back, about **3°** at its anterior end.
- **Riedel** dropped infratemporal-fossa registration in 1974 (the maxilla lengthens at PNS); **Baumrind** (31 implant subjects): ANS–PNS superimposition masks downward remodelling.
- **Nielsen** (18 implant subjects): palatal plane at ANS underestimates maxillary eruption by **30–50 %**; the Björk–Skieller method differed significantly from implants horizontally, not vertically.

✅ Number	Meaning
50 of ~700	patients usable
4.2 ± 0.96 y	mean interval between films
0.95 / 0.97 mm	anterior / posterior zygomatic drift
0.85 mm	internal palate: best number, rejected by eye
−5° to +8°	palatal-plane rotation range
−6° to +9°	S-N rotation against the implant line
1.5 : 1	orbital-floor rise : palatal-plane lowering
0.5 mm	total tracing error

Maxillary Superimposition: A Comparison of Three Methods for Cephalometric Evaluation of Growth and Treatment Change

by **Ib Leth Nielsen** · Am J Orthod Dentofacial Orthop 1989;95(5):422–431 · University of California, San Francisco

Study purpose

1. To compare three maxillary superimposition methods: **best fit, implant and structural**.
2. To test the **validity and reliability** of the **structural method**.

Background (pp 422–423):

- The traditional method superimposes along the **palatal plane (ANS–PNS), registered at ANS** (Broadbent 1937; Ricketts), and assumes the nasal floor does not change.
- Björk's implants showed **differential remodelling**. The nasal floor is lowered by resorption, usually more anteriorly, while the **anterior surface of the zygomatic process stays unchanged** except at key ridge and the orbital floor. Isaacson, Worms and Speidel (1976) found large molar and incisor differences between best fit and implants in 4 of Björk and Skieller's 21 cases.

Sample and methods

- **18 subjects** (6 boys, 12 girls) were usable from **35** implant subjects at UCSF. They were selected on film quality only: a clear zygomatic process, ANS and orbital floors. Films with excessive double contours were excluded.
- Age at T1 **10 ± 0.37 y**; interval **4 ± 0.56 y**. Treated and untreated subjects were mixed; most were Class II, some Class I, none Class III. The sample was too small to split by sex or treatment.
- Implants were placed by **open surgery**. Each subject needed at least **3–4 stable implants** in the zygomatic process on both films. Not every subject had implants on both sides.
- Each film was traced separately on mat acetate for each method. **Implants were left off** the best-fit and structural tracings to avoid bias. All three methods had **double determinations**, and a **second observer** repeated the structural method twice after brief instructions. Landmarks and registration points were digitised with the UCSF computer-aided head-film analysis (Baumrind and Miller), and the T2 − T1 displacements were computed (Tables I–II). The differences between methods are in Tables III–IV.

The three methods

Method	Superimposed on	Orientation	Status
Best fit	Optimal fit of the hard palate, nasal floors aligned	Registered at **ANS** and the ANS–A point region	Traditional; ignores nasal-floor resorption
Implant	Implants traced, distance between them bisected; unstable implants excluded	Implants	**Gold standard**
Structural (Björk–Skieller)	**Anterior surface of the zygomatic process**, tracing both zygomatic processes, orbital floors and palatal floor	T2 turned so orbital-floor apposition **equals** nasal-floor resorption	Anatomic method for patients without implants

- **Biological basis:** in Björk and Skieller's nine boys aged 4–20 y, orbital-floor apposition and nasal-floor resorption were in a ratio of about **3 : 2** (Fig 2). For this 4-year sample Nielsen set the two **equal (1 : 1)**.

"Palatal plane, registered at ANS": what it means

- **Orientation line:** the T2 ANS–PNS line is laid over the T1 line. This fixes the **rotation** between the films.
- **Pin point (registration):** ANS on T2 is placed on ANS on T1. This fixes the **translation**.
- **Why it under-reads eruption:** the palate is not a stable structure. The nasal floor sinks by resorption, more at ANS (−1.62 mm) than at PNS (−0.70 mm) on implants (Table I). Laying the T2 palate onto the T1 palate cancels that downward drift, so the descent of the teeth measured against the stable maxillary body is under-read by the same amount. Because the palate sinks unevenly, the T2 film is also rotated (the paper's en-masse rotational shift).

Landmarks and reference axes

Landmark	Definition (p 425)
ANS	Apex of the anterior nasal spine
ANS (Harvold)	Point on the inferior surface of the spine where the bone is 3 mm thick
PNS	Most posterior point of the hard palate, where the nasal floor meets the pterygomaxillary fissure
Point A	Most posterior point on the anterior contour of the maxilla
U6C / U6A	Mesiobuccal cusp / apex of the maxillary first molar
UIE / UIA	Incisal edge / apex of the most prominent maxillary central incisor

- **X axis = Downs' occlusal plane** of the T1 film (the line bisecting the first-molar and central-incisor occlusion); **Y axis = perpendicular through the first-molar mesial cusps** (Fig 6).
- Negative Y = downward (descent, eruption); negative X = posterior. The paper reads PNS X −2.34 as 2.3 mm backward.

Key results: vertical displacement over 4 years (Table II, mm)

Method	Molar cusp U6C	Molar apex U6A	Incisor edge UIE	Incisor apex UIA
Best fit	−3.40	−2.33	−1.40	−2.60
Implant	**−4.75**	**−3.80**	**−3.39**	**−4.36**
Structural	−4.99	−4.02	−3.39	−4.48

The printed table gives implant U6C as **−4.75 mm** (p 427), not −4.15. **Skeletal landmarks (Table I, mm, X / Y):**

Method	PNS	ANS	Point A
Best fit	−1.18 / +0.36	+0.62 / +0.28	+0.14 / −0.47
Implant	**−2.34 / −0.70**	**−0.63 / −1.62**	**−0.78 / −2.33**
Structural	−1.81 / −1.02	−0.12 / −1.72	−0.65 / −2.43

- **Best fit** showed the smallest vertical change. PNS, ANS and ANS (Harvold) moved slightly up (+0.36, +0.28, +0.37 mm). Point A dropped only **0.5 mm**, against **2.3 mm** on implants. The incisal edge descended about half as far as on implants (1.40 vs 3.39 mm).
- **Implant (Table I):** ANS −1.62, PNS −0.70, A point −2.33 mm. **ANS lowered about twice as much as PNS**, because nasal-floor resorption is greater anteriorly. ANS (Harvold) gave almost the same value (−1.56 mm).

↔️ Key results: horizontal displacement

Method	U6C X	U6A X	UIE X	UIA X
Best fit	+3.02	+0.92	+1.22	−0.08
Implant	+1.31	−0.42	−0.46	−1.17
Structural	+2.06	+0.22	+0.15	−0.86

- **PNS moved back 2.3 mm** on implants, against 1.2 mm with best fit. This is growth in maxillary length from sutural growth and tuberosity apposition, and best fit under-reads it by about **50 %**.
- **Point A:** 0.1 mm with best fit (SD ± 2.2, very variable) vs 0.8 mm backward with implants.
- **Structural vs implant:** every point is moved by a systematic **≈ 0.5 mm shift**. The Discussion links it to the zygomatic surface used for registration, which slopes distally and inferiorly, together with the slightly greater vertical displacement.

🧮 Statistical significance (Tables III–IV, mean differences, mm)

Comparison	Vertical	Horizontal
Implant – best fit	1.08–2.24 skeletal, 1.33–1.84 dental; all significant	1.09–1.69; all significant
Implant – structural	−0.04 to −0.33; none significant (all 8 negative, so structural reads slightly more descent)	0.41–0.66; 7 of 8 significant (Point A 0.41 NS)
Structural – best fit	1.39–2.00; all highly significant	0.63–1.07; 5 of 8 significant
Observers (structural)	ANS, ANS (Harvold), A point, UIE, UIA: 0.52–0.88	None significant

- **Best fit vs implant:** significantly different at every landmark (all P < .01; all but PNS horizontal P < .001). Best fit under-reads eruption by nearly 30 % at the molars and up to 50 % at the incisors.
- **Second observer (structural):** differences only in the vertical, at ANS, A point and the incisor points. The author puts this down to systematic observer differences and points that are hard to locate.
- The footnotes differ: Table III * P < .05, ** < .01, *** < .001; Table IV prints * < .05, ** < .001, *** < .0001.

🔎 Important observations

- **Nasal-floor modelling is greater anteriorly:** ANS dropped about twice as far as PNS with both the implant method (1.62 vs 0.70 mm, derived 2.3×) and the structural method (1.72 vs 1.02 mm, derived 1.7×).
- **Best fit causes an en-masse rotational shift** of the maxilla because it ignores the uneven lowering of the nasal floor. On average it differs from implants by **1.2 mm** horizontally across all points.
- **Anterior zygomatic surface:** implant superimposition showed no resorption here, contrary to Enlow, Kuroda and Lewis.
- **Variability** was large but similar in all three methods, so it came mainly from locating landmarks rather than from the superimposition. Implants on both sides in every subject might have reduced head-position error further.
- **Not tested (author):** the effects of treatment, sex and growth rate on remodelling, and changes in tooth inclination.

🦷 Clinical implications

- **Best fit:** not reliable for judging treatment outcome. It under-reads eruption and vertical growth, and about half the growth at PNS.
- **Implant:** assumed to be the most accurate, but it needs surgically placed implants, so it cannot be used in routine patients.
- **Structural:** the alternative for patients without implants; valid, especially vertically. It needs correct contrast and density, and the double contours of both zygomatic processes must be minimal or equal on all films (only 18 of 35 were usable). A short zygomatic process can add a rotation that misreads tooth movement.
- **Other registrations: PTM (Moore)** adds error because the maxilla lengthens at PNS. The **infratemporal fossa (Riedel)** has the same problem and also uses the posterior zygomatic surface, which grows by apposition. **Luder's key ridge** ignores remodelling and the lowering of key ridge by apposition, so it also under-reads eruption.

✅ Conclusions

1. The **structural method is valid and reliable**, especially for tooth eruption, but needs good films.
2. **Best fit** under-reads eruption by about 30 % at the molars and up to 50 % at the incisors.
3. Superimposing on the palatal plane at PTM, and Riedel's and Luder's methods, all introduce error.
4. If implants are taken as the most accurate method, the structural method is more valid than best fit.

✅ Number	Meaning
18 of 35	subjects usable
−3.40 / −4.75 / −4.99 mm	U6C vertical: best fit / implant / structural
−1.40 / −3.39 / −3.39 mm	UIE vertical: best fit / implant / structural
−1.62 / −0.70 / −2.33 mm	ANS / PNS / A point vertical with implants
≈ 30 % / ≈ 50 %	eruption missed by best fit: molars / incisors
2.3 vs 1.2 mm	PNS backward: implant vs best fit
≈ 0.5 mm	structural method's horizontal shift
3 : 2	orbital apposition : nasal resorption (Björk–Skieller, 4–20 y); 1 : 1 used here

Three-Dimensional Mandibular Regional Superimposition in Growing Patients

by **Nguyen, Cevidanes, Franchi, Ruellas and Jackson** · Am J Orthod Dentofacial Orthop 2018;153(5):747–754 · UNC Chapel Hill

📚 Study aims

1. Identify **stable mandibular structures in 3D** in growing patients, using **bone plates and screws** as reference, like Björk's implants (a "modified regional implant technique").
2. Test the **reproducibility** of 3D voxel-based mandibular regional superimposition on those structures.

Why it was needed (Introduction, pp. 747–749):

- In 2D the mandible is registered by **Björk's structural method**: the inner cortex of the inferior symphyseal border, the anterior chin above pogonion, the symphyseal trabecular pattern, and the canal trabeculae or the lower contour of a third-molar germ before root formation. These came from tantalum-implant studies.
- 2D flattens a 3D object and carries unavoidable errors: magnification, head position, landmark identification and tracing, and poorly reproducible reference lines or planes. No PA-cephalogram method exists for superimposing asymmetric growth.
- Earlier 3D mandibular work had gaps: **Koerich et al** (as cited) registered the whole mandible in non-growing patients; **Ruellas et al** compared regions with no external reference such as mini-implants.
- Implant studies cannot be repeated today (ALARA, ethics boards), so immobile **bone-anchor plates** served as the regional implants.

🔬 Materials and methods

	Aim 1: stability	Aim 2: reproducibility
Patients	**20** Class III, treated with bone anchors	**25** growing: 10 Class II Herbst, 10 Class II elastics, 5 Class III bone anchors
Age	**12.1 ± 1.3 y** at T1, **13.3 ± 1.1 y** at T2	**12.7 ± 1.4 y**; CVM stages 2–5
Interval	about 1 year	**12.6 ± 0.9 months**
Registered on	Plates and screws (the reference)	Chin + symphysis, by two observers
Outcome	Absolute mean surface error at each candidate region	Surface difference between the two observers' T2 models; ICC

- **Plates and screws** sat around the incisor apices, between the lateral incisor and canine (Fig 1), and were clinically immobile at T1 and T2.
- **Scans:** i-CAT or NewTom 3G, **0.3 mm voxels**, 20–25 s; T1 (pretreatment) and T2 on the same machine for each patient, drawn from earlier research and clinical databases.
- **Excluded:** craniofacial syndromes, significant asymmetric growth, and scans with double images from patient motion.
- **Sample size (aim 2):** calculated for a two-way random ICC (minimum acceptable 0.75 vs expected 0.95, α 0.05, power 0.80): 25 patients, 2 examiners.
- **Error metric:** absolute mean surface error, the average size of the gaps over the whole region regardless of direction. Chosen over root mean square, which weights outliers more heavily. Normality checked with Kolmogorov–Smirnov.

🔀 How the 3D registration is done

1. **Orient:** for aim 2, observer 1 oriented T1 to **Frankfort horizontal** in a known 3D coordinate system.
2. **Approximate:** T2 is first placed roughly over T1 on the analogous anatomy.
3. **Register:** a validated, automated **voxel-based registration** in Slicer CMF (version 3.1) on the chosen region only (the plates in aim 1, chin + symphysis in aim 2). It uses thousands of voxels rather than a few landmarks or traced lines.
4. **Segment and model:** after registration, mandibles segmented by automated thresholding, refined in ITK-SNAP (version 3.6), and converted to 3D surface models.
5. **Measure:** surface distances between T1 and T2 models, shown as colour-coded distance maps and overlays (Figs 3–8).

Regional vs overall superimposition: an *overall* (cranial-base) registration, as in Cevidanes' earlier 3D work, shows where the mandible moved relative to the skull. A *regional* registration fixes the mandible on its own stable bone, so what remains is **remodelling inside the mandible and tooth movement within it**: the 3D version of Björk's mandibular superimposition. The region chosen decides the answer, which is why each candidate region was tested first.

📍 Regions evaluated (Fig 2)

1. **Chin:** anterior surface from pogonion up to B point, laterally to the distal-incisal points of the right and left lateral incisors.
2. **Symphysis:** inner (lingual) cortex, bounded laterally at the limit of its lingual surface, from the inferior border up to the level of the centres of both mental foramina.
3. **Third-molar crypts:** inferior contour of the third-molar germs, a **candidate stable region from Björk's 2D list**.

- **Mandibular canal:** excluded **a priori**, because Björk (1963) and later Krarup et al (2005) found that it changes with growth.

📊 Key results: registration error against the bone plates (Table, p. 749)

Region	Mean error, mm	SD	P	Reading
Chin	**0.37**	0.16	.765	**Stable**, submillimetre
Symphysis	**0.40**	0.14	.698	**Stable**, almost identical to the chin (text prints SD 0.15)
Third-molar crypt	**1.94**	0.06	.0025	**Unstable**, significant

Chin and symphysis are stable in growing patients; the third-molar crypt is not. The crypt error was about **5 ×** the chin error (derived) and, per the Discussion (p. 753), was worst in patients who grew vertically. The test behind the P values is not named.

🔁 Reproducibility across observers (aim 2)

- Because chin and symphysis were equally stable, they were **combined into one registration field**: more surface gives the software more to match, as Björk used several structures together in 2D, and Ruellas et al found larger regions gave smaller errors.
- Two observers registered the 25 pairs independently; their T2 models were overlaid.
- **Interobserver error 0.12 ± 0.11 mm** (Results; the abstract misprints ± 1.1), below the 0.3 mm voxel.
- ICC 0.998 (95 % CI 0.995–1.000).
- **One example patient (Figs 7–8):** difference under **0.25 mm** over the whole mandible and within **0.12 mm** over **75 %** of its surface.

🦷 Key observations and visual findings (Figs 4–6)

Area	Change seen	Agrees with
Anterior chin contour at pogonion	**Little or no remodelling**	Björk; Björk and Skieller
Inferior border of the symphysis	Bone **deposition** or remodelling, about **2 mm** near menton (Fig 4D)	Björk; Björk and Skieller
Posterior-inferior border of the mandible	Resorption (as printed; the paper cites Fig 4, which shows only the symphysis)	Björk; Björk and Skieller
Anterior border of the ascending ramus	**Resorption** (Fig 6)	Enlow and Harris
Posterior border of the ramus	Bone **deposition** (Fig 6)	Enlow and Harris
Third-molar crypt	Large displacement, mostly vertical (Fig 5)	Ruellas et al: 16 of 18 superimpositions failed with the crypts included (as cited)

The obvious objection, answered by the authors (p. 753): are the chin and symphysis "stable" only because the plates sit next to them?

- If so, menton should fit too, yet Fig 4D (captioned as a symphysis registration) shows about **2 mm** of deposition there while the chin and inner symphysis fit within about 0.3 mm.
- If a small plate error were simply magnified with distance, the posterior border near the antegonial notch would be as wrong as the crypt. Fig 5 (one colour map, not measured) shows the crypt error is larger.

💡 Clinical and research implications

- **3D voxel-based superimposition overcomes 2D limits:** no magnification error; head-position error corrected easily without losing information; thousands of voxels instead of a few traced landmarks or lines; automated software reduces operator bias.
- **Register on chin + symphysis** to see true changes within the growing mandible in all three planes, including asymmetry and structures a cephalogram hides, such as the condyles.
- **Leave out the third-molar crypt and the canal** in 3D, although both are in Björk's 2D list.
- **Research:** a reproducible reference for 3D growth-modification studies. It was reproducible in Herbst, elastics and bone-anchor patients, but stability was tested against plates in Class III patients only.

⚠️ Limitations (our reading; the paper lists none)

- About one year of growth, small samples, all treated; aim 1 is Class III only.
- Two observers agreeing shows reproducibility, not accuracy; the accuracy evidence rests on the 20 plate patients.
- Chin and symphysis errors are about **1.2–1.3 voxels** (derived), not below the voxel size.

🧠 Conclusions

1. The **chin and symphysis** are stable regions for 3D mandibular regional superimposition in growing patients.
2. Registering on them **combined is reliable and reproducible** between observers.
3. From the Results and Discussion: unlike Björk's 2D list, the **third-molar crypt** is unstable in 3D (1.94 mm); the **canal** was excluded in advance because Björk himself found it changes with growth.
4. From the Discussion: 3D removes magnification error, lets head-position error be corrected easily, and uses thousands of voxels instead of a few landmarks.

✅ Number	Meaning
20 / 25	patients for stability / reproducibility
0.37 / 0.40 mm	chin / symphysis error against the plates
1.94 mm, P = .0025	third-molar crypt error
0.12 ± 0.11 mm	interobserver error, chin + symphysis
0.998 (0.995–1.000)	ICC between two observers
0.3 mm	voxel size
≈ 2 mm	deposition or remodelling near menton (Fig 4D)
12.6 ± 0.9 months	reproducibility scan interval